Introduction to Machine Learning and Retrieval-Augmented Generation

A Short Technical Reference for a Local PDF Question-Answering Demonstration

1. What Is Machine Learning?

Machine learning is a branch of artificial intelligence that enables computer systems to identify patterns in data and use those patterns to make predictions or decisions. Instead of writing a separate rule for every situation, developers train a model using examples. The model adjusts its internal parameters to reduce the difference between predictions and expected results.

2. Common Types of Machine Learning

Supervised learning uses labelled examples. Each training example contains input data and a known target. Classification predicts a category, such as whether an email is spam. Regression predicts a numerical value, such as a house price.

Unsupervised learning

Unsupervised learning works with data that does not have labelled target values. It can discover groups, similarities, or unusual patterns. Clustering is a common unsupervised learning task.

Reinforcement learning

Reinforcement learning trains an agent to choose actions through interaction with an environment. The agent receives rewards or penalties and learns a strategy that aims to maximize long-term reward.

3. Training, Loss, and Gradient Descent

During training, a model produces predictions for input examples. A loss function measures how far those predictions are from the expected answers. The training process attempts to minimize this loss. Gradient descent is an optimization method used to update model parameters. It moves parameters in the direction that reduces loss. The learning rate controls the size of each update.

4. Overfitting and Regularization

Overfitting occurs when a model learns details and noise specific to its training data instead of learning patterns that generalize to new examples. An overfit model may perform very well on training data but poorly on unseen data. Regularization discourages unnecessarily complex models. Other approaches include collecting representative data, using validation data, applying early stopping, and reducing model complexity.

5. Neural Networks and Deep Learning

A neural network is made of layers of interconnected computational units. Each unit combines input values with learned weights, applies an activation function, and passes a result to the next layer. Deep learning uses neural networks with multiple layers to learn complex representations. Convolutional neural networks are often used for image tasks. Recurrent neural networks process sequences, and long short-term memory networks are designed to retain useful information over longer sequences.

6. Text Embeddings and Vector Search

A text embedding is a numerical vector that represents the meaning or features of a piece of text. An embedding model maps text into a vector space. Texts with related meanings may have vectors that are close under a chosen distance or similarity measure. A vector database stores embeddings together with associated text and metadata. Similarity search converts a user query into an embedding and retrieves stored vectors that are close to it.

7. Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation combines information retrieval with a language model. A RAG system first searches a knowledge collection for relevant passages. It then provides those passages as context to the language model, which generates an answer based on the retrieved material. A typical pipeline loads source documents, splits them into chunks, creates embeddings, stores chunks in a vector database, retrieves relevant chunks for a question, and passes the question plus context to a language model.

8. Document Chunking and Retrieval Quality

Chunking divides a long document into smaller sections that can be embedded and retrieved independently. Chunk size affects how much context is available for each match. Overlap preserves some text between adjacent chunks so that information near a boundary is less likely to be lost. Retrieval quality depends on source documents, chunk size, embedding model, distance metric, and number of retrieved results.

9. Grounded Answer Generation and Limitations

A RAG prompt should instruct the language model to answer using the supplied context and to say when the context does not contain the answer. This can reduce unsupported claims, but it does not guarantee correctness. RAG systems can fail when documents contain poor text extraction, relevant passages are not retrieved, or the language model misinterprets the context. Evaluation should examine both retrieval results and final answers.

10. Key Terms

Embedding: a numerical representation of text. Vector database: a system for storing and searching vectors. Chunk: a smaller segment of a larger document. Similarity search: retrieval based on closeness in vector space. Metadata: descriptive fields stored with a document. Retriever: the component that selects relevant documents. Prompt: instructions and context supplied to a language model. Grounded answer: an answer supported by retrieved source material.

End of reference document.